House Price Prediction – Project Summary

Objective:

The goal of this project was to build machine learning models to predict house prices using property features such as area, bedrooms, bathrooms, stories, parking availability, and housing amenities.

Data Preparation:

The housing dataset was explored and cleaned. No missing values or duplicate records were found. Categorical variables such as main road access, guest room availability, basement, air conditioning, preferred area, and furnishing status were converted into numerical format using one-hot encoding.

Model Development:

Two regression models were trained and evaluated: Linear Regression and Random Forest Regressor. The dataset was split into training and testing sets using an 80/20 ratio.

Results:

Linear Regression achieved the best performance with an R^2 score of approximately 0.65, outperforming the Random Forest model. The model was able to explain about 65% of the variation in house prices and produced reasonably accurate predictions.

Key Findings:

Property area, number of bathrooms, bedrooms, parking spaces, and housing amenities were among the most influential factors affecting house prices. Larger homes with better facilities generally had higher prices.

Recommendation:

Real estate businesses should prioritize property size and key amenities when estimating market value and advising buyers or sellers. Data-driven pricing strategies can improve valuation accuracy and support better decision-making.